Week 5 Level 3: Advanced

Sorting, Stacks and Queues

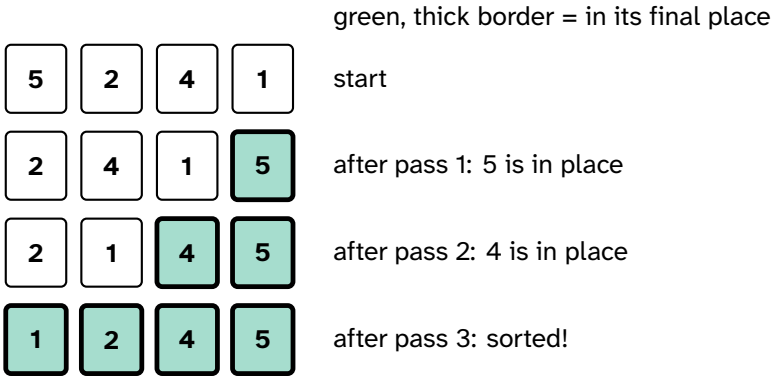
Sorting puts data in order; a stack remembers what to undo; a queue makes sure everyone gets a turn.

What you will learn

- ☐ Sort a list with bubble sort and selection sort, and count the swaps.
- ☐ Sort a leaderboard with `sorted(key=..., reverse=True)`.
- ☐ Use a stack to undo, and a queue to take turns.

Two ways to sort

Selection sort: find the smallest item and swap it to the front. Then find the smallest of the rest and swap it into second place. Repeat.
Bubble sort: walk along the list comparing neighbors. If the left one is bigger, swap them. After each pass the biggest item has "bubbled" to the end.



Python's own sort

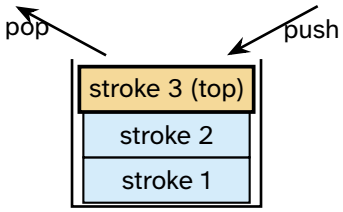
Python can sort for you. The **key** is a function that says *what to sort by*:

```
scores = [("Ada", 120), ("Leo", 95), ("Mei", 150)]

def points_of(entry):
```

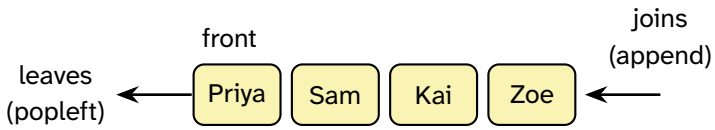
```
    return entry[1]                    # sort by the number

best_first = sorted(scores, key=points_of, reverse=True)
print(best_first)   # [('Mei', 150), ('Ada', 120), ('Leo', 95)]
```

`sorted` makes a **new** list and leaves the old one alone.

Stacks and queues



Stack: last in, first out



Queue: first in, first out

A **stack** is like a pile of plates: you can only take the top one. “Undo” uses a stack: the last thing you did is the first thing undone. A **queue** is like a lunch line: join at the back, leave from the front. In Python, `collections.deque` is a fast queue.

Exercise 5.1: Sort Race

Folder: `week05/ex-sort-race`. `selection_sort` is done. Finish the swap inside `bubble_sort`:

```
if result[i] > result[i + 1]:
    # swap both at once:
    result[i], result[i + 1] = result[i + 1], result[i]
    swaps = swaps + 1
```

Swaps for the 8 cards: selection _____ bubble _____

Exercise 5.2: Leaderboard

Folder: `week05/ex-leaderboard`. Finish `name_of` (sort A to Z, ignoring capitals), `top_scores` and the fair tie rule in `rank_lines`: equal points share a place, so the board reads 1, 2, 3, 3, 5.

Exercise 5.3: Undo Paint

Folder: `week05/ex-undo-paint`. Finish the `Stack` class: `push`, `pop`, `peek` and `size`. Then run it, click the window, draw with the arrow keys and press u to undo. Each stroke is one item on the stack.

Tip: Kind leaderboards

A good leaderboard uses nicknames, lets ties share a place, and celebrates personal bests and “most improved”, not only the winner.

Vocabulary box

sort Put items in order, for example smallest to largest.

swap Two items change places.

key function A function that tells `sorted` what to sort by.

stack A collection where the last item in is the first out (push and pop).

queue A collection where the first item in is the first out (join at the back, leave at the front).

deque Python’s fast queue, from `collections`. Say “deck”.

Reflection

Where do you see a stack and a queue in real life or in apps you use?

Why is “undo” a stack and not a queue?

Week 5
Sorting, Stacks and Queues
Level 3: Advanced

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Turn Queue (about 10 minutes)

Four friends share one game. A queue makes it fair: the player at the front plays, then goes to the back of the line.

1. Open `week05/hw-turn-queue/starter.py` in Thonny.
2. Finish `take_turn`: `player = queue.popleft()`, then `queue.append(player)`, then return the player.
3. Finish `leave`: remove a player who has to go home, and return `True` (or `False` if they were not in the queue).
4. Run it, then open `check_my_work.py` and press Run. Commit your work in `git-quest`.

Predict: the queue is Priya, Sam, Kai, Zoe. Who plays turns 1 to 6? _____

★★ Challenge (want more?): Stack or queue?

Write **S** (stack) or **Q** (queue) next to each one, and say why in a few words.

Situation	S or Q	Why
The back button in a web browser		
Songs waiting in a playlist		
Printing 3 documents sent one after another		
A pile of homework sheets, newest on top		

Creative (make it yours): Design a fair leaderboard

Sketch a leaderboard screen for a game you like. Use nicknames only. Include at least one thing that celebrates effort, not only winning (for example “most improved”, “personal best”, or “longest streak

of helping”).

Self-check: tick each box when it is true

- ☐ All checks pass in `hw-turn-queue`.
- ☐ I can explain “last in, first out” and “first in, first out”.
- ☐ My leaderboard sketch uses nicknames only.
- ☐ I committed my work in `git-quest`.